

Showcase

Mine-FNL

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Real outputs from the v0.1 release on synthetic contracts.

Fixture

We generate two PDFs (acme_msa.pdf + nda.pdf) using reportlab, ingest them, and ask four questions. The answers below are the actual outputs of qwen2.5-coder:7b running locally on an M4 — no human editing.

Q1: "What's the cap on liability in the MSA?"

```
$ firm-bot query demo "What's the cap on liability in the MSA?"
```

```
answer: "The cap on liability in the MSA is the fees paid by Acme Corp to  
        Demo LLP in the twelve (12) months preceding the event giving  
        rise to the claim. [acme_msa.pdf:p.1]"
```

```
cited: ["acme_msa.pdf:p.1"]
```

```
hits:
```

```
[1] score=0.0164 bm25_rank=0 dense_rank=0  
    acme_msa.pdf:p.1  
    "Section 4.2 – Limitation of Liability\nEXCEPT FOR BREACHES OF ..."  
[2] score=0.0161 bm25_rank=1 dense_rank=1  
    acme_msa.pdf:p.1 ← continuation chunk from same page  
[3] score=0.0087 bm25_rank=-1 dense_rank=2  
    acme_msa.pdf:p.1 ← Section 6.1 Indemnification
```

What went right: the model cited exactly the right clause. Both BM25 and dense retrieval agreed on the rank-0 hit, so the fusion boosted it above neighbouring chunks. The operator could click [acme_msa.pdf:p.1] in the UI and see Section 4.2 in the source pane.

What could go wrong: a smaller model (1.5B) might paraphrase without attaching the marker. The guard model would flag that as unsupported.

Q2: "How long does the NDA's confidentiality last?"

```
answer: "The NDA's confidentiality obligations remain in effect for  
        three (3) years from the Effective Date, and confidentiality
```

```
obligations survive termination for an additional five (5)
years. [nda.pdf:p.1]"
cited: ["nda.pdf:p.1"]
hits:
[1] score=0.0089 bm25_rank=0 dense_rank=0
nda.pdf:p.1
"Term\nThis Agreement shall remain in effect for three (3) years
from the Effective Date. Confidentiality obligations shall
survive termination for an additional five (5) years."
```

What went right: the model hit both clauses of the answer (the 3-year term AND the 5-year survival), each within a single cited source. The user can verify by clicking the citation chip — the source pane shows the exact text the model quoted.

Q3: "What are the hourly rates?"

```
answer: "The hourly rates per Schedule A are: Senior Partner $650/hour,
Partner $475/hour, Senior Associate $325/hour, Associate
$210/hour, Paralegal $125/hour. [acme_msa.pdf:p.1]"
cited: ["acme_msa.pdf:p.1"]
hits:
[1] score=0.0085 bm25_rank=-1 dense_rank=0
acme_msa.pdf:p.1
"Schedule A — Fee Schedule\nHourly rates: Senior Partner
$650/hour, Partner $475/hour, Senior Associate $325/hour,
Associate $210/hour, Paralegal $125/hour."
```

What went right: the model listed all five tiers verbatim from Schedule A. The retrieval hit the right chunk even though BM25 didn't match ("rates" vs. the structured header "Schedule A — Fee Schedule"); the dense embedding bridged the lexical gap.

Q4: "What's our exposure if Demo LLP breaches confidentiality?"

This question requires *connecting* two clauses — the 12-month-fees cap (Section 4.2) with the carve-out (also Section 4.2). The structurally-aware chunker keeps Section 4.2 in a single chunk so both clauses are retrieved together.

```
answer: "Based on Section 4.2 — Limitation of Liability, breaches of
confidentiality or indemnification obligations are carved out
of the 12-month-fees cap. That is, indemnity exposure is
uncapped. [acme_msa.pdf:p.1]"
cited: ["acme_msa.pdf:p.1"]
hits:
[1] score=0.0164 bm25_rank=0 dense_rank=0
acme_msa.pdf:p.1
"Section 4.2 — Limitation of Liability\nEXCEPT FOR BREACHES OF
```

CONFIDENTIALITY OR INDEMNIFICATION OBLIGATIONS, NEITHER PARTY'S
AGGREGATE LIABILITY ... SHALL EXCEED THE FEES PAID ..."

Why this is interesting: a naive chunker that splits Section 4.2 in half would only retrieve one of the two clauses, and the model would say "12 months of fees" without mentioning the carve-out. The structure-aware chunker keeps both clauses together — the model sees the exception, and the answer reflects it.

Reproducing this

```
git clone https://github.com/firm-bot/firm-bot
cd firm-bot
pip install -e .
```

```
ollama pull qwen2.5-coder:14b      # answer model
ollama pull qwen2.5-coder:7b      # judge model
```

```
PYTHONPATH=. python examples/make_sample_firm.py data/firms/demo/source
firm-bot firm create --slug demo --name "Demo LLP"
firm-bot ingest demo
firm-bot query demo "What's the cap on liability in the MSA?"
```

The first ingest takes ~30 s (downloads the embedding model). Subsequent queries on the same firm are sub-second for retrieval and ~5–15 s for the answer.